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(Bnous) Another Example of Confirmatory Factor Analysis

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```
library(lavaan)
library(semPlot)
library(semptools)
library(lavaangui)
library(dplyr)
library(semTools)
library(modelsummary)
library(tinytable)
```

```
asags <- read.csv("https://osf.io/download/sy2d4/")
```

Dataset: Attainment of School Achievement Goal Scale (A-SAGS)

The **A-SAGS** (Boivin et al., 2026) measures the perceived attainment of three types of achievement goals in academic settings. Participants rated 13 items on a **7-point Likert scale** (1 = not at all to 7 = completely).

Factor	Items	Description
MGA (Mastery Goal Attainment)	ga1, ga4, ga7, ga10, ga13	Learning and developing competence
SRGA (Social Reference Goal Attainment)	ga2, ga5, ga8, ga11	Outperforming others
NGA (Nurturance Goal Attainment)	ga3, ga6, ga9, ga12	Helping others succeed

The three subscales are hypothesized to reflect a single higher-order construct: **Goal Attainment (GA)**. The dataset has N = 527 with some missing values.

Descriptive Statistics

```
modelsummary::datasummary_skim(asags)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
ga1	8	1	4.1	1.5	1.0	4.0	7.0	
ga2	8	2	3.2	1.6	1.0	3.0	7.0	
ga3	8	1	3.0	1.6	1.0	3.0	7.0	
ga4	8	1	4.1	1.5	1.0	4.0	7.0	
ga5	8	2	3.1	1.6	1.0	3.0	7.0	
ga6	8	2	3.2	1.6	1.0	3.0	7.0	
ga7	8	2	4.1	1.5	1.0	4.0	7.0	
ga8	8	1	3.1	1.6	1.0	3.0	7.0	
ga9	8	2	2.9	1.6	1.0	3.0	7.0	
ga10	8	1	3.5	1.5	1.0	4.0	7.0	
ga11	8	1	3.0	1.6	1.0	3.0	7.0	
ga12	8	2	3.1	1.6	1.0	3.0	7.0	
ga13	8	1	4.0	1.6	1.0	4.0	7.0	

Inter-item Correlations

```
cor_asags <- cor(asags, use = "pairwise.complete.obs")
bg <- rgb(as.integer(cor_asags > 0), 0, as.integer(cor_asags < 0),
          alpha = abs(cor_asags))
bg[upper.tri(cor_asags, diag = TRUE)] <- NA
```

```
nv <- ncol(cor_asags)
asags |>
  modelsummary::datasummary_correlation() |>
  style_tt(i = 1:nv, j = 1:nv + 1, background = bg)
```

	ga1	ga2	ga3	ga4	ga5	ga6	ga7	ga8	ga9	ga10	ga11	ga12	ga13
ga1	1	.	.	.	.	.	.	.	.	.	.	.	.
ga2	.48	1	.	.	.	.	.	.	.	.	.	.	.
ga3	.39	.65	1	.	.	.	.	.	.	.	.	.	.
ga4	.72	.53	.47	1	.	.	.	.	.	.	.	.	.
ga5	.45	.87	.65	.53	1	.	.	.	.	.	.	.	.
ga6	.42	.67	.85	.50	.68	1	.	.	.	.	.	.	.
ga7	.66	.50	.45	.77	.51	.52	1	.	.	.	.	.	.
ga8	.40	.82	.58	.46	.83	.60	.43	1	.	.	.	.	.
ga9	.47	.68	.79	.53	.68	.82	.53	.68	1	.	.	.	.
ga10	.55	.63	.60	.63	.62	.65	.63	.63	.75	1	.	.	.
ga11	.43	.83	.64	.50	.85	.66	.48	.84	.75	.68	1	.	.
ga12	.40	.64	.79	.48	.64	.89	.49	.62	.86	.70	.70	1	.
ga13	.59	.57	.51	.66	.56	.58	.68	.55	.61	.76	.57	.61	1

> **ℹ Note**
>
> Blue cells indicate positive correlations, red cells negative, intensity reflects magnitude. Items within the same subscale (MGA: ga1, ga4, ga7, ga10, ga13; SRGA: ga2, ga5, ga8, ga11; NGA: ga3, ga6, ga9, ga12) should cluster together.

Dimensionality

Before imposing a factor structure via CFA, it is useful to get a rough sense of how many dimensions underlie the 13 items. We inspect **eigenvalues** of the correlation matrix and run a quick **parallel analysis**.

```
fa_parallel <- psych::fa.parallel(
  asags,
  n.iter = 100, show.legend = FALSE, main = ""
)
```

```
Parallel analysis suggests that the number of factors =  3  and the number
of components =  2
```

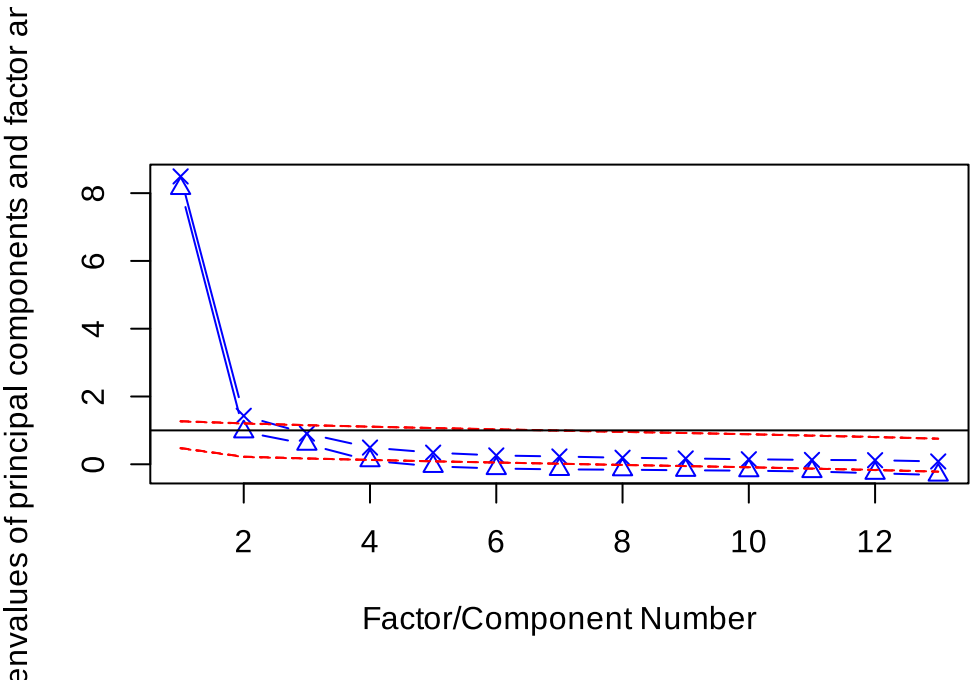


Figure 1: Scree plot with parallel analysis reference line. Factors above the red line are retained.

```
ev <- eigen(cor_asags, only.values = TRUE)$values
data.frame(
  Factor        = seq_along(ev),
  Eigenvalue    = round(ev, 3),
  Cumulative_var = round(cumsum(ev) / sum(ev) * 100, 1)
)
```

```
   Factor Eigenvalue Cumulative_var
1       1      8.494           65.3
2       2      1.431           76.3
3       3      0.905           83.3
4       4      0.493           87.1
5       5      0.343           89.7
6       6      0.260           91.7
7       7      0.230           93.5
8       8      0.193           95.0
9       9      0.172           96.3
10     10      0.148           97.4
11     11      0.129           98.4
12     12      0.120           99.4
13     13      0.083          100.0
```

> **Note**
>
> Parallel analysis retains factors whose eigenvalues exceed those expected from random data. The eigenvalue-greater-than-1 rule (Kaiser criterion) tends to over-extract and should be treated as an upper bound only.

The A-SAGS was designed with **3 first-order factors** plus an overarching **goal attainment** dimension, so we expect 1–3 meaningful factors. We proceed to CFA to evaluate this structure formally.

Confirmatory Factor Analysis (Chapter 13)

Confirmatory Factor Analysis (CFA) is a measurement model where we specify how observed variables (indicators) relate to underlying latent constructs (factors). Unlike EFA, CFA requires a priori specification of the factor structure.

Key Concepts

- **Latent Variables (Factors):** Unobserved constructs (e.g., goal attainment) that cause observed scores.
- **Indicators:** Observed variables (e.g., survey items) used to measure the latent construct.
- **Factor Loadings (λ):** Regression coefficients representing the strength of the relationship between the factor and the indicator.
- **Measurement Errors (ϵ):** The portion of the indicator variance not explained by the factor.

Identification in CFA

To estimate a CFA model, we must set the scale for the latent variable. Two common methods:

1. **Marker Variable:** Fix one factor loading to 1.0 (default in `lavaan`).
2. **Standardized Latent Variable:** Fix the factor variance to 1.0 (used here with `std.lv = TRUE`).

A Note on Estimation

The A-SAGS items are technically ordinal, but with 7 response categories they can reasonably be treated as approximately continuous. We use **MLR** (Maximum Likelihood with Robust standard errors), which:

- Treats items as continuous (appropriate for ≥ 5 ordered categories)
- Provides Satorra–Bentler scaled χ^2 and robust standard errors that are robust to non-normality
- Produces AIC and BIC for model comparison

> **ℹ Note**
>
> For fewer response categories (≤ 4), or when the distribution is highly skewed, prefer `ordered = TRUE` with `estimator = "WLSMV"` instead.

Example 1: First-Order CFA (3 Factors)

We first fit a correlated three-factor model to evaluate the measurement structure of each subscale before introducing the higher-order factor.

```
cfa_model <- "
  MGA  =~ ga1 + ga4 + ga7 + ga10 + ga13
  SRGA =~ ga2 + ga5 + ga8 + ga11
  NGA  =~ ga3 + ga6 + ga9 + ga12
"

fit_cfa <- cfa(cfa_model, data = asags,
               std.lv    = TRUE,
               estimator = "MLR")

summary(fit_cfa, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 32 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        29

                                                   Used       Total
  Number of observations                            492         527

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               530.098     403.225
  Degrees of freedom                                62          62
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.315
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                              7050.570    4549.389
  Degrees of freedom                                78          78
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.550

User Model versus Baseline Model:
```

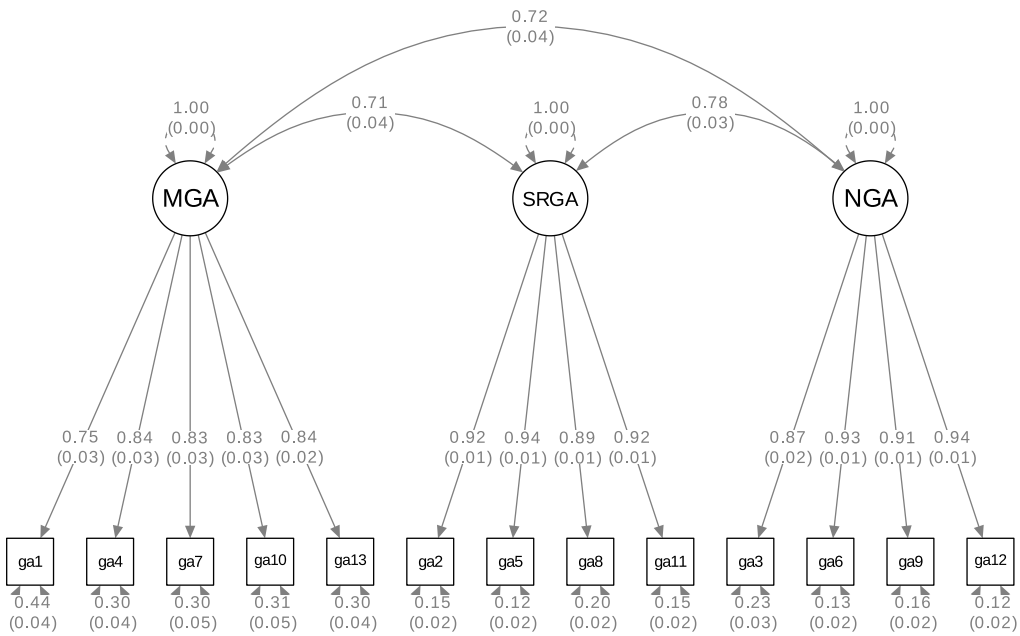
```
  Comparative Fit Index (CFI)                    0.933       0.924
  Tucker-Lewis Index (TLI)                       0.916       0.904

  Robust Comparative Fit Index (CFI)                         0.935
  Robust Tucker-Lewis Index (TLI)                            0.919

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -8706.090   -8706.090
  Scaling correction factor                                  1.678
      for the MLR correction
  Loglikelihood unrestricted model (H1)      -8441.041   -8441.041
  Scaling correction factor                                  1.430
      for the MLR correction

  Akaike (AIC)                               17470.180   17470.180
  Bayesian (BIC)                             17591.936   17591.936
  Sample-size adjusted Bayesian (SABIC)      17499.890   17499.890

Root Mean Square Error of Approximation:

  RMSEA                                          0.124       0.106
  90 Percent confidence interval - lower         0.114       0.097
  90 Percent confidence interval - upper         0.134       0.114
  P-value H_0: RMSEA <= 0.050                    0.000       0.000
  P-value H_0: RMSEA >= 0.080                    1.000       1.000

  Robust RMSEA                                               0.121
  90 Percent confidence interval - lower                     0.110
  90 Percent confidence interval - upper                     0.133
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         1.000

Standardized Root Mean Square Residual:

  SRMR                                           0.056       0.056

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  MGA =~
    ga1               1.095    0.064   17.239    0.000    1.095    0.746
```

```
    ga4               1.274    0.059   21.705    0.000    1.274    0.838
    ga7               1.255    0.065   19.350    0.000    1.255    0.835
    ga10              1.239    0.062   20.079    0.000    1.239    0.828
    ga13              1.313    0.058   22.698    0.000    1.313    0.837
  SRGA =~
    ga2               1.496    0.046   32.645    0.000    1.496    0.919
    ga5               1.528    0.044   34.570    0.000    1.528    0.937
    ga8               1.411    0.048   29.190    0.000    1.411    0.892
    ga11              1.494    0.045   33.059    0.000    1.494    0.920
  NGA =~
    ga3               1.364    0.053   25.967    0.000    1.364    0.875
    ga6               1.515    0.045   33.779    0.000    1.515    0.934
    ga9               1.485    0.051   29.059    0.000    1.485    0.914
    ga12              1.516    0.047   32.195    0.000    1.516    0.938

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  MGA ~~
    SRGA              0.707    0.036   19.513    0.000    0.707    0.707
    NGA               0.724    0.035   20.494    0.000    0.724    0.724
  SRGA ~~
    NGA               0.780    0.029   27.086    0.000    0.780    0.780

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .ga1               0.958    0.093   10.348    0.000    0.958    0.444
   .ga4               0.686    0.101    6.803    0.000    0.686    0.297
   .ga7               0.686    0.103    6.670    0.000    0.686    0.304
   .ga10              0.705    0.099    7.123    0.000    0.705    0.315
   .ga13              0.737    0.092    8.048    0.000    0.737    0.299
   .ga2               0.410    0.054    7.625    0.000    0.410    0.155
   .ga5               0.323    0.045    7.216    0.000    0.323    0.121
   .ga8               0.512    0.056    9.181    0.000    0.512    0.205
   .ga11              0.405    0.060    6.705    0.000    0.405    0.154
   .ga3               0.571    0.074    7.769    0.000    0.571    0.235
   .ga6               0.334    0.065    5.106    0.000    0.334    0.127
   .ga9               0.433    0.052    8.273    0.000    0.433    0.164
   .ga12              0.311    0.042    7.395    0.000    0.311    0.119
    MGA               1.000                               1.000    1.000
    SRGA              1.000                               1.000    1.000
    NGA               1.000                               1.000    1.000
```

```
p <- semPaths(fit_cfa, whatLabels = "std", edge.label.cex = 0.8,
              style = "ram", nCharNodes = 0, mar = c(5, 1, 12, 1))
```

```
std_fit <- standardizedSolution(fit_cfa)
names(std_fit)[names(std_fit) == "est.std"] <- "est"
```

```
p |>
  mark_se(ests = std_fit, sep = "\n") |>
  set_cfa_layout(fcov_curve = 2.75, loading_position = .7) |>
  plot()
```



Example 2: Bi-factor CFA

The **bi-factor model** is an alternative to the hierarchical model for representing a general factor alongside specific factors. Instead of routing the general factor through the subscales, the bi-factor model has:

- A **general factor (GA)** loading directly on all 13 items
- Three **specific factors** (MGA, SRGA, NGA) each loading on their respective items
- All factors constrained to be **orthogonal** (uncorrelated)

This allows the model to partition item variance into general and specific components simultaneously, rather than assuming the general factor operates solely through the subscale factors.

In `lavaan`, bi-factor models are specified with `bifactor()` inside `cfa()`:

```
cfa_bi <- "
  GA   =~ ga1 + ga2 + ga3 + ga4 + ga5 + ga6 +
          ga7 + ga8 + ga9 + ga10 + ga11 + ga12 + ga13
  MGA  =~ ga1 + ga4 + ga7 + ga10 + ga13
  SRGA =~ ga2 + ga5 + ga8 + ga11
  NGA  =~ ga3 + ga6 + ga9 + ga12
"
```

```
fit_bi <- cfa(cfa_bi, data = asags,
              std.lv     = TRUE,
              estimator  = "MLR",
              orthogonal = TRUE)

summary(fit_bi, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 39 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        39

                                                   Used       Total
  Number of observations                           492         527

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               257.207     211.560
  Degrees of freedom                                52          52
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.216
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                              7050.570    4549.389
  Degrees of freedom                                78          78
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.550

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.971       0.964
  Tucker-Lewis Index (TLI)                       0.956       0.946

  Robust Comparative Fit Index (CFI)                         0.972
  Robust Tucker-Lewis Index (TLI)                            0.958

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -8569.644   -8569.644
  Scaling correction factor                                  1.716
      for the MLR correction
  Loglikelihood unrestricted model (H1)      -8441.041   -8441.041
  Scaling correction factor                                  1.430
      for the MLR correction
```

```
  Akaike (AIC)                                17217.289   17217.289
  Bayesian (BIC)                              17381.029   17381.029
  Sample-size adjusted Bayesian (SABIC)       17257.244   17257.244

Root Mean Square Error of Approximation:

  RMSEA                                           0.090       0.079
  90 Percent confidence interval - lower          0.079       0.069
  90 Percent confidence interval - upper          0.101       0.089
  P-value H_0: RMSEA <= 0.050                     0.000       0.000
  P-value H_0: RMSEA >= 0.080                     0.929       0.446

  Robust RMSEA                                                0.087
  90 Percent confidence interval - lower                      0.075
  90 Percent confidence interval - upper                      0.099
  P-value H_0: Robust RMSEA <= 0.050                          0.000
  P-value H_0: Robust RMSEA >= 0.080                          0.837

Standardized Root Mean Square Residual:

  SRMR                                            0.033       0.033

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  GA =~
    ga1               0.748    0.070   10.665    0.000    0.748    0.509
    ga2               1.215    0.068   17.872    0.000    1.215    0.747
    ga3               1.158    0.072   16.134    0.000    1.158    0.742
    ga4               0.890    0.072   12.418    0.000    0.890    0.586
    ga5               1.222    0.066   18.517    0.000    1.222    0.750
    ga6               1.275    0.063   20.080    0.000    1.275    0.786
    ga7               0.871    0.069   12.666    0.000    0.871    0.579
    ga8               1.190    0.064   18.735    0.000    1.190    0.752
    ga9               1.450    0.063   22.989    0.000    1.450    0.893
    ga10              1.272    0.059   21.599    0.000    1.272    0.849
    ga11              1.333    0.058   22.863    0.000    1.333    0.821
    ga12              1.372    0.062   22.109    0.000    1.372    0.849
    ga13              1.147    0.076   15.157    0.000    1.147    0.731
  MGA =~
    ga1               0.873    0.067   12.956    0.000    0.873    0.594
    ga4               1.018    0.062   16.504    0.000    1.018    0.669
```

```
    ga7               0.990    0.068   14.493    0.000    0.990    0.659
    ga10              0.330    0.077    4.275    0.000    0.330    0.221
    ga13              0.587    0.096    6.115    0.000    0.587    0.374
  SRGA =~
    ga2               0.873    0.079   11.006    0.000    0.873    0.537
    ga5               0.934    0.073   12.803    0.000    0.934    0.573
    ga8               0.760    0.074   10.291    0.000    0.760    0.481
    ga11              0.688    0.074    9.345    0.000    0.688    0.424
  NGA =~
    ga3               0.738    0.096    7.703    0.000    0.738    0.473
    ga6               0.884    0.062   14.191    0.000    0.884    0.546
    ga9               0.407    0.122    3.325    0.001    0.407    0.250
    ga12              0.637    0.093    6.880    0.000    0.637    0.394

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  GA ~~
    MGA               0.000                               0.000    0.000
    SRGA              0.000                               0.000    0.000
    NGA               0.000                               0.000    0.000
  MGA ~~
    SRGA              0.000                               0.000    0.000
    NGA               0.000                               0.000    0.000
  SRGA ~~
    NGA               0.000                               0.000    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .ga1               0.835    0.091    9.210    0.000    0.835    0.387
   .ga2               0.408    0.062    6.618    0.000    0.408    0.154
   .ga3               0.547    0.074    7.445    0.000    0.547    0.225
   .ga4               0.483    0.067    7.193    0.000    0.483    0.209
   .ga5               0.292    0.052    5.670    0.000    0.292    0.110
   .ga6               0.221    0.079    2.791    0.005    0.221    0.084
   .ga7               0.521    0.095    5.489    0.000    0.521    0.231
   .ga8               0.509    0.056    9.122    0.000    0.509    0.204
   .ga9               0.371    0.044    8.330    0.000    0.371    0.141
   .ga10              0.515    0.072    7.115    0.000    0.515    0.230
   .ga11              0.387    0.049    7.877    0.000    0.387    0.147
   .ga12              0.321    0.039    8.225    0.000    0.321    0.123
   .ga13              0.801    0.085    9.370    0.000    0.801    0.326
    GA                1.000                               1.000    1.000
    MGA               1.000                               1.000    1.000
    SRGA              1.000                               1.000    1.000
    NGA               1.000                               1.000    1.000
```

```
lavaangui::plot_lavaan(fit_bi)
```

ℹ Hierarchical vs. Bi-factor

	Hierarchical	Bi-factor
General factor	Acts through subscales	Acts directly on items
Specific factors	Residual variance of subscales	Residual variance of items after removing general factor
Factor correlations	General ↔ specific: fixed (via structure)	All factors orthogonal
df	More constrained	Same or fewer constraints

The bi-factor model is more appropriate when you believe the general factor directly influences item responses, not just indirectly through group factors.

Model Fit

Correlation Residuals

The factor model assumes that the latent factors fully explain the correlations among the indicators. Residuals > |0.10| suggest local misfit.

```
res_cor <- lavResiduals(fit_bi, type = "cor")$cov
bg_res <- rgb(as.integer(res_cor > 0), 0, as.integer(res_cor < 0),
              alpha = abs(res_cor))
bg_res[upper.tri(res_cor, diag = TRUE) | abs(res_cor) <= 0.1] <- NA

nv_res <- ncol(res_cor)
res_df <- as.data.frame(round(res_cor, 3))
tt(cbind(" " = rownames(res_df), res_df)) |>
  style_tt(i = 1:nv_res, j = 1:nv_res + 1, background = bg_res)
```

	ga1	ga2	ga3	ga4	ga5	ga6	ga7	ga8	ga9	ga10	ga11	ga12	ga13
ga1	0.000	0.086	0.024	0.017	0.072	0.029	−0.011	0.014	0.013	−0.007	0.004	−0.022	−0.014
ga2	0.086	0.000	0.100	0.085	0.006	0.083	0.066	−0.004	0.007	−0.002	−0.007	0.009	0.020
ga3	0.024	0.100	0.000	0.041	0.089	0.006	0.011	0.023	0.019	−0.031	0.032	−0.023	−0.032
ga4	0.017	0.085	0.041	0.000	0.105	0.051	0.004	0.019	0.008	−0.016	0.020	−0.011	−0.019
ga5	0.072	0.006	0.089	0.105	0.000	0.090	0.075	−0.006	0.016	−0.008	−0.002	−0.001	0.023
ga6	0.029	0.083	0.006	0.051	0.090	0.000	0.060	0.010	−0.013	−0.013	0.021	0.004	0.009
ga7	−0.011	0.066	0.011	0.004	0.075	0.060	0.000	−0.016	0.005	−0.004	−0.005	0.001	0.007
ga8	0.014	−0.004	0.023	0.019	−0.006	0.010	−0.016	0.000	0.007	−0.005	0.015	−0.018	0.002

	ga1	ga2	ga3	ga4	ga5	ga6	ga7	ga8	ga9	ga10	ga11	ga12	ga13
ga9	0.013	0.007	0.019	0.008	0.016	−0.013	0.005	0.007	0.000	−0.009	0.013	0.014	−0.035
ga10	−0.007	−0.002	−0.031	−0.016	−0.008	−0.013	−0.004	−0.005	−0.009	0.000	−0.009	−0.015	0.064
ga11	0.004	−0.007	0.032	0.020	−0.002	0.021	−0.005	0.015	0.013	−0.009	0.000	0.004	−0.025
ga12	−0.022	0.009	−0.023	−0.011	−0.001	0.004	0.001	−0.018	0.014	−0.015	0.004	0.000	−0.005
ga13	−0.014	0.020	−0.032	−0.019	0.023	0.009	0.007	0.002	−0.035	0.064	−0.025	−0.005	0.000

Modification Indices

Modification indices (MIs) estimate how much the overall χ^2 would decrease if a currently fixed parameter were freed.

```
modindices(fit_bi, sort. = TRUE, minimum.value = 10)
```

```
     lhs op  rhs     mi    epc sepc.lv sepc.all sepc.nox
150 ga10 ~~ ga13 44.197  0.246   0.246    0.382    0.382
128  ga6 ~~  ga9 35.039 -0.167  -0.167   -0.582   -0.582
107  ga3 ~~ ga12 35.039 -0.218  -0.218   -0.519   -0.519
49  SRGA ~~  NGA 32.107  0.502   0.502    0.502    0.502
118  ga5 ~~  ga6 23.308  0.097   0.097    0.382    0.382
65  SRGA =~ ga12 23.265 -0.181  -0.181   -0.112   -0.112
61  SRGA =~  ga6 18.762  0.166   0.166    0.103    0.103
52   MGA =~  ga5 16.493  0.144   0.144    0.088    0.088
124  ga5 ~~ ga12 15.969 -0.079  -0.079   -0.259   -0.259
70   NGA =~  ga5 15.073  0.157   0.157    0.096    0.096
147  ga9 ~~ ga13 14.107 -0.119  -0.119   -0.218   -0.218
146  ga9 ~~ ga12 14.101  0.092   0.092    0.266    0.266
101  ga3 ~~  ga6 14.100  0.231   0.231    0.665    0.665
72   NGA =~  ga8 14.064 -0.173  -0.173   -0.109   -0.109
47   MGA ~~ SRGA 13.934  0.263   0.263    0.263    0.263
53   MGA =~  ga6 13.721  0.131   0.131    0.081    0.081
59  SRGA =~  ga3 12.628  0.162   0.162    0.104    0.104
73   NGA =~ ga10 11.718 -0.201  -0.201   -0.134   -0.134
68   NGA =~  ga2 11.059  0.144   0.144    0.089    0.089
130  ga6 ~~ ga11 10.841 -0.069  -0.069   -0.237   -0.237
56   MGA =~ ga11 10.820 -0.123  -0.123   -0.076   -0.076
131  ga6 ~~ ga12 10.381  0.163   0.163    0.612    0.612
104  ga3 ~~  ga9 10.381  0.087   0.087    0.193    0.193
```

Goodness of Fit Indices

Exact Fit

- **Model Chi-Square (χ^2)**
  - Tests the null hypothesis that the model perfectly reproduces the population covariance matrix.

- **Limitation**: Highly sensitive to sample size; rejection does not indicate the degree of misfit.

Approximate Fit Indices

- **RMSEA**: Misfit per degree of freedom, adjusted for model complexity.
  Cutoffs: < 0.05 (close), 0.05–0.08 (reasonable), > 0.10 (poor).
  Includes 90% CI and test of close fit (H_0: RMSEA ≤ 0.05).
- **CFI**: Compares the target model to the independence (null) baseline model.
  ≥ 0.95 indicates good fit; less affected by sample size than χ^2.
- **SRMR**: Average standardized residual between observed and model-implied correlations.
  < 0.08 indicates acceptable fit.

> ℹ Note
>
> **Robust Fit Indices with MLR**
>
> With `estimator = "MLR"`, lavaan computes several versions of each index:
>
> | Suffix | Based on | Recommendation |
> |---|---|---|
> | (none) | Naive ML χ^2 | Avoid; ignores non-normality |
> | `.scaled` | Satorra–Bentler scaled χ^2 | Common but suboptimal |
> | `.robust` | Yuan–Bentler T_2^* | **Preferred** |
>
> Savalei & Rosseel recommend the `.robust` variants because the Yuan–Bentler T_2^* statistic yields better-behaved fit indices under non-normality than the Satorra–Bentler correction. In lavaan these are `cfi.robust`, `tli.robust`, `rmsea.robust`, etc.

> ⚠ Warning
>
> **Cutoff Values Should Not Be Used Mechanically**
>
> - Cutoffs are rough guidelines, not absolute rules.
> - Consider sample size, model complexity, and estimation method.
> - A model with "good" fit indices can be theoretically meaningless.
> - Always examine residuals, modification indices, and parameter estimates.

Comparing Models

The first-order and bi-factor models are **not nested**, so we compare them using information criteria (AIC/BIC) rather than a χ^2 difference test:

```
lavTestLRT(fit_cfa, fit_bi)   # approximate; use AIC/BIC for non-nested
comparison
```

```
Scaled Chi-Squared Difference Test (method = "satorra.bentler.2001")

lavaan->lavTestLRT():
   lavaan NOTE: The "Chisq" column contains standard test statistics, not the
   robust test that should be reported per model. A robust difference test is
    a function of two standard (not robust) statistics.

        Df   AIC   BIC  Chisq Chisq diff   RMSEA Df diff Pr(>Chisq)
fit_bi  52 17217 17381 257.21
fit_cfa 62 17470 17592 530.10     149.22 0.22748      10  < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Extracting Fit Indices

```
fitMeasures(fit_bi, c("chisq", "df", "pvalue",
                      "cfi.robust", "tli.robust",
                                   "rmsea.robust",  "rmsea.ci.lower.robust",
"rmsea.ci.upper.robust",
                      "srmr", "aic", "bic"))
```

```
                chisq                    df                pvalue
              257.207                52.000                 0.000
           cfi.robust            tli.robust          rmsea.robust
                0.972                 0.958                 0.087
rmsea.ci.lower.robust rmsea.ci.upper.robust                  srmr
                0.075                 0.099                 0.033
                  aic                   bic
            17217.289             17381.029
```

Quick Table of Results

```
# performance::model_performance() only extracts naive (non-robust) indices.
# glance_custom.lavaan appends the robust variants so gof_map can reference
them.
glance_custom.lavaan <- function(x, ...) {
  m <- lavaan::fitmeasures(x)
  tibble::tibble(
    cfi.robust   = m[["cfi.robust"]],
    tli.robust   = m[["tli.robust"]],
    rmsea.robust = m[["rmsea.robust"]]
  )
}

gm <- data.frame(
  raw   = c("nobs", "statistic", "chi2.df", "p.chi2",
```

```
          "cfi.robust", "tli.robust", "rmsea.robust", "srmr", "aic", "bic"),
  clean = c("N", "Chisq", "df", "p(Chisq)",
            "CFI*", "TLI*", "RMSEA*", "SRMR", "AIC", "BIC"),
  fmt   = c(0, 2, 0, 3, 3, 3, 3, 3, 1, 1)
)
modelsummary::msummary(
  list("First-order" = fit_cfa, "Bi-factor" = fit_bi),
  fmt      = fmt_decimal(digits = 2),
  estimate = "{estimate} [{conf.low}, {conf.high}]",
  statistic = NULL,
  metrics  = "all",
  gof_map  = gm
)
```

	First-order	Bi-factor
MGA = ga1	1.10 [0.97, 1.22]	0.87 [0.74, 1.01]
MGA = ga4	1.27 [1.16, 1.39]	1.02 [0.90, 1.14]
MGA = ga7	1.25 [1.13, 1.38]	0.99 [0.86, 1.12]
MGA = ga10	1.24 [1.12, 1.36]	0.33 [0.18, 0.48]
MGA = ga13	1.31 [1.20, 1.43]	0.59 [0.40, 0.77]
SRGA = ga2	1.50 [1.41, 1.59]	0.87 [0.72, 1.03]
SRGA = ga5	1.53 [1.44, 1.62]	0.93 [0.79, 1.08]
SRGA = ga8	1.41 [1.32, 1.51]	0.76 [0.62, 0.90]
SRGA = ga11	1.49 [1.41, 1.58]	0.69 [0.54, 0.83]
NGA = ga3	1.36 [1.26, 1.47]	0.74 [0.55, 0.93]
NGA = ga6	1.51 [1.43, 1.60]	0.88 [0.76, 1.01]
NGA = ga9	1.48 [1.38, 1.59]	0.41 [0.17, 0.65]
NGA = ga12	1.52 [1.42, 1.61]	0.64 [0.46, 0.82]
MGA SRGA	0.71 [0.64, 0.78]	0.00 [0.00, 0.00]
MGA NGA	0.72 [0.65, 0.79]	0.00 [0.00, 0.00]
SRGA NGA	0.78 [0.72, 0.84]	0.00 [0.00, 0.00]
GA = ga1		0.75 [0.61, 0.89]
GA = ga2		1.22 [1.08, 1.35]
GA = ga3		1.16 [1.02, 1.30]
GA = ga4		0.89 [0.75, 1.03]
GA = ga5		1.22 [1.09, 1.35]
GA = ga6		1.27 [1.15, 1.40]

	First-order	Bi-factor
GA = ga7		0.87 [0.74, 1.01]
GA = ga8		1.19 [1.07, 1.31]
GA = ga9		1.45 [1.33, 1.57]
GA = ga10		1.27 [1.16, 1.39]
GA = ga11		1.33 [1.22, 1.45]
GA = ga12		1.37 [1.25, 1.49]
GA = ga13		1.15 [1.00, 1.30]
GA MGA		0.00 [0.00, 0.00]
GA SRGA		0.00 [0.00, 0.00]
GA NGA		0.00 [0.00, 0.00]
N	492	492
Chisq	530.10	257.21
df	62	52
p(Chisq)	0.000	0.000
CFI*	0.935	0.972
TLI*	0.919	0.958
RMSEA*	0.121	0.087
SRMR	0.056	0.033
AIC	17470.2	17217.3
BIC	17591.9	17381.0

(Predicted) Factor Scores and Diagnostics

```
fs <- lavPredict(fit_bi)
head(fs)
```

```
             GA        MGA         SRGA         NGA
[1,] -1.1136729 -0.4005389 -0.736901059  0.22125627
[2,]  1.2255373 -0.1511204  0.420705357  0.32106284
[3,]  0.2326275 -1.1541345  0.400258127  0.23580948
[4,]  0.2712853 -0.3597672  0.004817669  0.60267246
[5,] -1.5011042 -1.1057333 -0.234199877 -0.20409779
[6,] -0.8160404 -1.0705456 -0.074889489 -0.07796274
```

Residuals: Indicator – Predicted Indicator Scores

```
resid_bi <- lavInspect(fit_bi, "data") - lavPredict(fit_bi, type = "ov")
psych::pairs.panels(resid_bi)
```

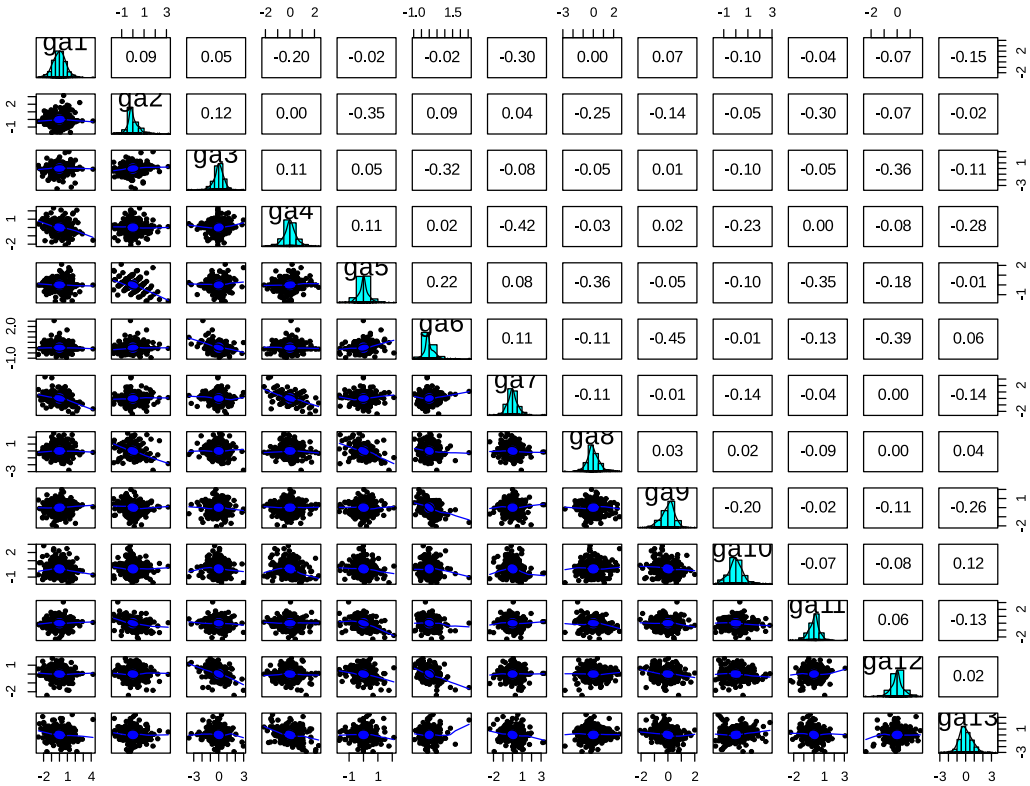


Figure 2: Indicator residuals from the bi-factor CFA model.

Composite Reliability

```
semTools::compRelSEM(fit_cfa)  # first-order model (subscale reliability)
```

```
$MGA

Composite `MGA` is composed of observed variables:
    ga1, ga4, ga7, ga10, ga13
True-score variance is represented by common factor(s):
    MGA
Total variance of composite `MGA` determined from the unrestricted model.
The proportion attributable to "true" scores is its model-based estimate of
reliability ("omega"):

[1] 0.909

$SRGA
```

```
Composite `SRGA` is composed of observed variables:
    ga2, ga5, ga8, ga11
True-score variance is represented by common factor(s):
    SRGA
Total variance of composite `SRGA` determined from the unrestricted model.
The proportion attributable to "true" scores is its model-based estimate of
reliability ("omega"):

[1] 0.955

$NGA

Composite `NGA` is composed of observed variables:
    ga3, ga6, ga9, ga12
True-score variance is represented by common factor(s):
    NGA
Total variance of composite `NGA` determined from the unrestricted model.
The proportion attributable to "true" scores is its model-based estimate of
reliability ("omega"):

[1] 0.955
```

References

Boivin, T., Veilleux, C., Egerton-Graham, M., Caron, P.-O., & Gaudreau, P. (2026). A tutorial on evaluating confirmatory factor analysis with lavaan and dynamic: Integrating statistical and conceptual approaches. PsyArXiv. https://doi.org/10.31234/osf.io/2m6rk_v4

Kline, R. B. (2023). Principles and practice of structural equation modeling (5th ed.). Guilford Press.